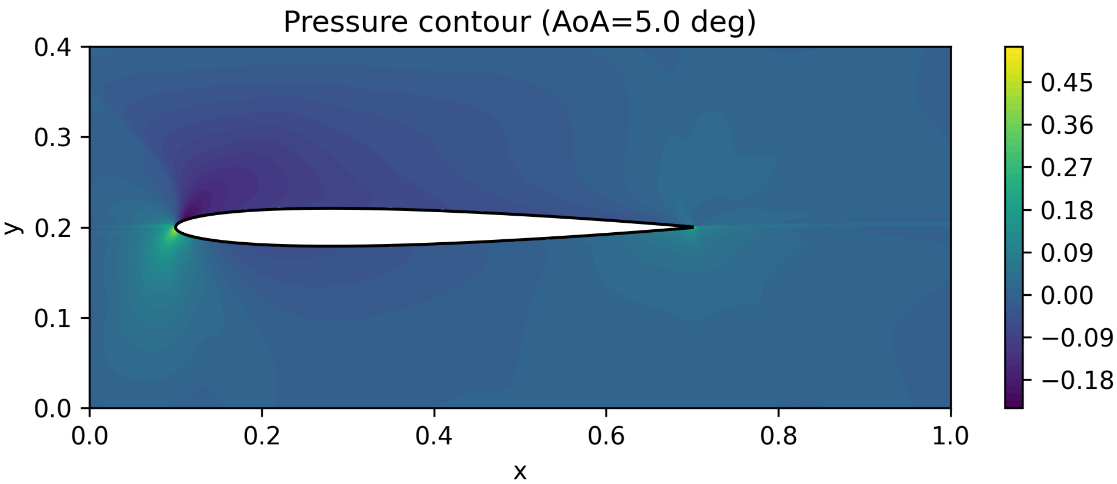
Airfoil Flow

Physics-Informed Neural Networks and MATLAB Potential-Flow Modeling



Predicted pressure field for the NACA 0009 case at a 5° angle of attack.

PROJECT	Two-dimensional incompressible flow over NACA airfoils
METHODS	PyTorch PINNs · MATLAB source-distribution potential flow
AUTHOR	Ali Sakhaei
REPORT	English portfolio edition · 2026

1. Project Overview

This project examines steady, two-dimensional flow around airfoils using two complementary numerical approaches. Physics-informed neural networks (PINNs) reconstruct velocity and pressure fields by minimizing governing-equation and boundary-condition residuals. A separate MATLAB model uses a regularized source distribution to study inviscid potential flow, numerical independence, and airfoil-thickness sensitivity.

Objectives

- Model incompressible flow without a conventional body-fitted CFD mesh.
- Compare a cambered NACA 23018 case at zero incidence with a symmetric NACA 0009 case at 5°.
- Estimate aerodynamic forces from the predicted pressure and stress fields.
- Evaluate numerical convergence and interpret limitations rather than treating every output as physically exact.
- Compare PINN behavior with a lower-cost MATLAB potential-flow formulation.

Cases and methods

Case	Geometry	Condition	Method
A	NACA 23018	$\alpha = 0°$, Re = 100	Viscous PINN
B	NACA 0009	$\alpha = 5°$	Potential-flow PINN
C	NACA 0009	$\alpha = 0°$, $U_\infty = 1$	MATLAB source distribution

Scope note

The work is an academic numerical investigation. It demonstrates model construction, physics-based loss design, post-processing, and critical validation. It is not presented as a certified aerodynamic design tool.

2. Governing Equations and PINN Formulation

For the viscous case, the network approximates the flow variables as functions of spatial coordinates. Automatic differentiation supplies the derivatives required by the steady incompressible Navier–Stokes equations.

$$\nabla \cdot \mathbf{u} = 0$$

$$(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u}$$

The training objective combines interior physics residuals with boundary constraints. A generic representation is:

$$\mathcal{L} = w_f \mathcal{L}_{PDE} + w_b \mathcal{L}_{boundary} + w_g \mathcal{L}_{gauge} + w_K \mathcal{L}_{Kutta}$$

Network design

The NACA 23018 model maps the two inputs (x, y) through five hidden layers of 50 neurons and predicts six field quantities: velocity, pressure, and stress components. The implementation also examines depth and width sensitivity using depths of 4, 6, 8, and 10 layers and widths of 20, 50, 80, and 100 neurons.

The final NACA 0009 model adds several stabilizing elements: far-field velocity constraints without over-constraining pressure, a pressure-gauge point, robust trailing-edge selection, a Kutta constraint on tangential velocity, and a learnable circulation parameter.

Aerodynamic coefficients

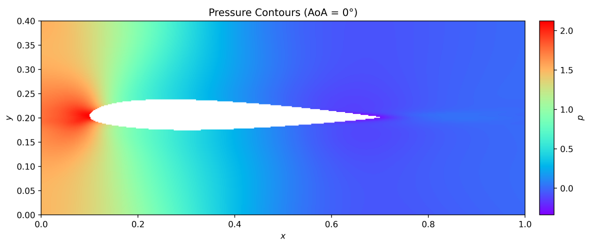
Surface traction is integrated around the airfoil. Lift and drag are then normalized by the free-stream dynamic pressure and chord:

$$C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 c} \qquad C_D = \frac{D}{\frac{1}{2}\rho U_\infty^2 c}$$

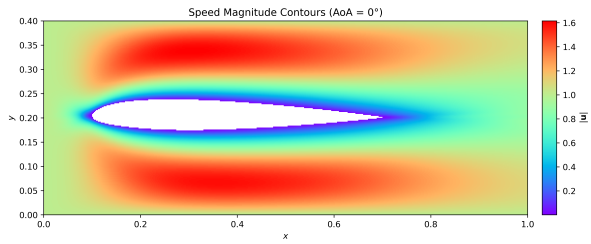
> This step is especially sensitive to surface ordering, normals, stress prediction, scaling, and numerical quadrature; therefore coefficient accuracy is assessed separately from visual field quality.

3. Case A — NACA 23018 at α = 0°

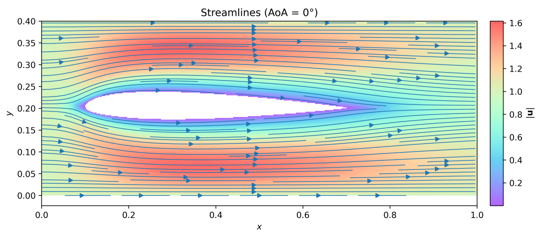
The first PINN solves a low-Reynolds-number flow problem for a cambered NACA 23018 airfoil. The predicted pressure, speed, and streamlines remain smooth and recover a localized disturbance around the body.



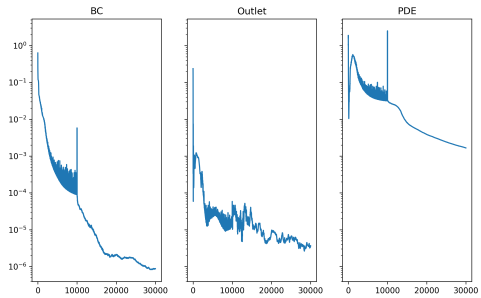
Pressure field



Speed magnitude



Streamlines



Training-loss history

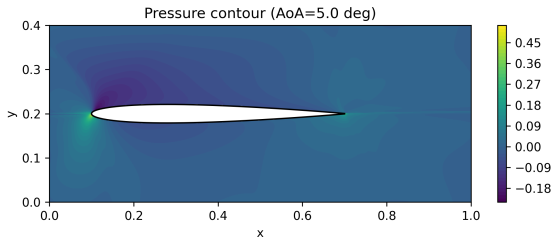
Reported force coefficients

Quantity	PINN result	Interpretation
C_L	0.29789	Close to the project's reference value of approximately 0.30
C_D	2.97713	Unphysically high; the report expects roughly 0.8–1.2 at Re = 100

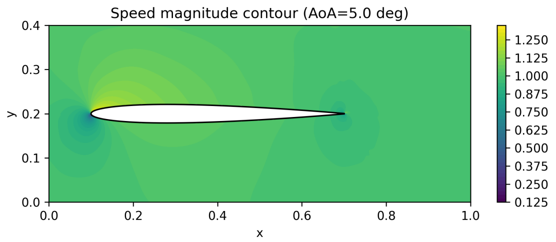
The lift estimate is encouraging, but the drag estimate is not accepted as a reliable physical prediction. Likely causes include surface-normal errors, stress uncertainty, scaling, and force-integration sensitivity. This discrepancy is retained as a documented limitation rather than hidden.

4. Case B — NACA 0009 at α = 5°

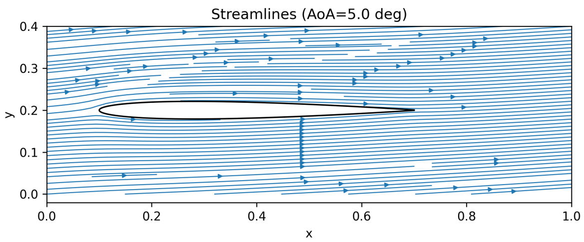
The second model targets potential flow over a symmetric NACA 0009 profile at positive incidence. The final implementation uses a learnable circulation term and a trailing-edge Kutta constraint to obtain the expected asymmetric pressure and velocity fields.



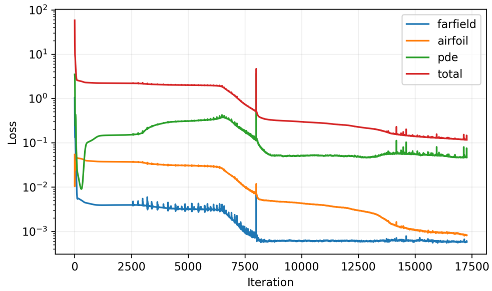
Pressure field



Speed magnitude



Streamlines



Training-loss history

Integrated results

L	D	C_L	C_D
4.625497×10^{-2}	4.530762×10^{-3}	0.1541832	0.0151025

Positive lift is qualitatively consistent with the 5° angle of attack. The small nonzero drag should be interpreted as a numerical residual, because ideal potential flow predicts zero pressure drag. The PDE loss also remains imperfectly converged, so the results are best used for qualitative and methodological comparison.

5. MATLAB Potential-Flow Study

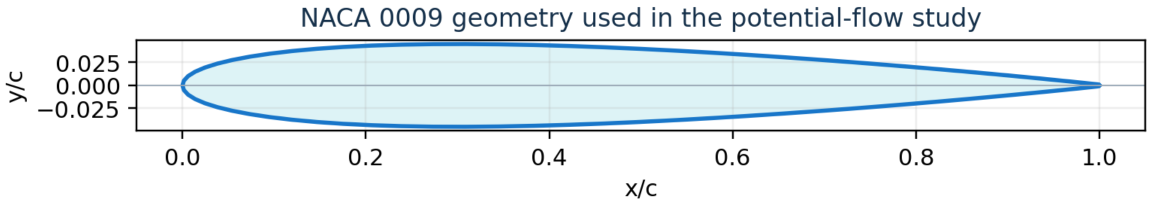
A separate MATLAB script models potential flow around the symmetric NACA 0009 airfoil using multiple rows of regularized point sources. Source strengths are obtained through weighted, regularized least squares while enforcing the no-penetration condition.

$$\phi = U_\infty (x\cos\alpha + y\sin\alpha) + \sum_{j=1}^{N} \frac{q_j}{2\pi} \ln r_j$$

$$\mathbf{V} = \nabla\phi \qquad \mathbf{V}\cdot\mathbf{n} \approx 0 \text{ on the surface}$$

Numerical-independence study

Target source count	Maximum |V·n|	Observation
60	1.22×10^{-1}	Boundary condition not yet sufficiently resolved
480	9.09×10^{-3}	Approximately enters the project’s trusted range



Airfoil coordinates supplied with the final project archive

Across the symmetric zero-incidence cases, lift remains near zero. Drag approaches a small residual on the order of 10^{-2}, consistent qualitatively with d’Alembert’s paradox but not identically zero because of discretization, source placement, and regularization. The thickness sweep is therefore interpreted as numerical sensitivity—not physical viscous drag.

6. Conclusions and Reproducibility

Main conclusions

- PINNs can reconstruct smooth pressure, speed, and streamline fields while embedding conservation laws directly in the training objective.
- The NACA 23018 model predicts lift close to the selected reference but substantially overpredicts drag, revealing the sensitivity of force extraction to surface stresses and integration.
- The corrected NACA 0009 model produces positive lift at positive incidence and a small residual drag consistent with numerical error in an ideal-flow formulation.
- The MATLAB source-distribution method reaches a lower no-penetration residual as the source count increases and provides a computationally simpler comparison model.
- The strongest result of the project is the full workflow: physics formulation, neural-network implementation, numerical post-processing, convergence assessment, and transparent error analysis.

Repository contents

Folder	Contents
notebooks/	Clean final notebooks for the NACA 23018 and NACA 0009 PINN cases
matlab/	Commented potential-flow and sensitivity-analysis script
data/	Airfoil coordinate files
models/	Saved PyTorch parameter files
results/	Loss histories, flow-field images, and MATLAB SVG figures
docs/	English portfolio report and original Persian report

Reproduction notes

Install the Python dependencies listed in requirements.txt, load the matching coordinate data, and run each notebook independently. Training a PINN can be computationally expensive; pretrained parameter files and exported result figures are included so the reported outcomes remain inspectable without retraining. For MATLAB, load the NACA 0009 coordinate matrix as **data** before running potential-flow-airfoil.m.

Known limitations

The available scripts do not constitute a validated CFD solver. Future improvements should include independent CFD or experimental validation, more systematic nondimensionalization, adaptive collocation, explicit uncertainty reporting, and stronger verification of surface-normal and force-integration routines.